

Han Zhou

han-zhou-math.github.io

Email: hzhou24@sas.upenn.edu

Mobile: +1 (267) 979-5378

EMPLOYMENT

-
- **University of Pennsylvania** Philadelphia, PA, USA
• *AMCS/CMB Postdoctoral Fellow, Department of Mathematics* Sep. 2024 -

EDUCATION

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- **Shanghai Jiao Tong University** Shanghai, PRC
• *Ph.D., Mathematics, Advisor: Wenjun Ying.* Sept. 2020 - Jun. 2024
 - **Shanghai Jiao Tong University** Shanghai, PRC
• *B.S., Mathematics and Applied Mathematics* Sept. 2016 - Jun. 2020

RESEARCH INTERESTS

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- **Numerical PDEs:** develop efficient numerical methods for complex and moving interface problems and related convergence analysis; Cartesian-grid methods (KFBIM, IIM, etc.), boundary integral methods
 - **Biophysical Modeling & Simulation:** membranes dynamics in viscous fluids; cell migration driven by passive or active stress
 - **PDE Analysis:** study fluid-structure interaction PDEs, including well-posedness, stability, bifurcation, etc.

PAPERS AND PREPRINTS

[†]Authors listed in alphabetical order. *Corresponding author.

1. **Han Zhou.** Open Inextensible Filaments in Planar Stokes Flow: Well-Posedness, Endpoint Asymptotics, and Straightening. Submitted. arXiv.
2. **Han Zhou,** Yoichiro Mori, Lingxing Yao. A Cartesian grid method for advection-diffusion equations with Robin boundary conditions on moving domains. Submitted. arXiv.
3. Eduardo García-Juárez, Susanna V. Haziot, Po-Chun Kuo, Yoichiro Mori, **Han Zhou.**[†] On the global asymptotic stability for the 3D Peskin Problem at critical regularity. Submitted. arXiv.
4. Sho Kawakami, **Han Zhou,** Po-Chun Kuo, Yoichiro Mori, Yuan-Nan Young. Stability and equilibria of a compressible elastic membrane in Stokes flow. Submitted. arXiv.
5. Shuo Ling, Wenjun Ying, **Han Zhou**^{*}. Principal-part decomposition for neural operator learning of Dirichlet-to-Neumann maps. Submitted. arXiv.
6. **Han Zhou,** Shuwang Li, Wenjun Ying. A Cartesian grid-based boundary integral method for moving interface problems. Submitted. arXiv.
7. **Han Zhou,** Minsheng Huang, Wenjun Ying. ADI schemes for the heat equation on arbitrary 3D domains and their applications. Submitted. arXiv.
8. Charles L. Epstein, Yoichiro Mori, **Han Zhou**^{*†}. Bulk-surface coupled PDE with an open boundary. *Studies in Applied Mathematics*, 157(2):e70286, 2026. DOI; arXiv.
9. **Han Zhou,** Wenjun Ying. A correction function-based KFBIM method for Brinkman interface problems. *Journal of Computational Physics*, 568:115404, 2027. DOI; arXiv.
10. **Han Zhou,** Yuan-Nan Young, Yoichiro Mori. Modeling and simulation of open membranes in Stokes flow with mixed-dimensional coupling. *Multiscale Modeling & Simulation*, 24(2):474–500, 2026. DOI; arXiv.
11. Pengsong Yin, Wenjun Ying, Yulin Zhang, **Han Zhou**^{*†}. A kernel-free boundary integral method for elliptic interface problems on surfaces. *Journal of Computational Physics*, 562:115004, 2026. DOI.
12. **Han Zhou,** Shuwang Li, Wenjun Ying. An alternating direction implicit method for mean curvature flows. *Journal of Scientific Computing*, 101:65, 2024. DOI.

13. **Han Zhou**, Wenjun Ying. A correction function-based kernel-free boundary integral method for elliptic PDEs with implicitly defined interfaces. *Journal of Computational Physics*, 496:112545, 2024. DOI.
14. **Han Zhou**, Jiahe Yang, Wenjun Ying. A kernel-free boundary integral method for the nonlinear Poisson-Boltzmann equation. *Journal of Computational Physics*, 493:112423, 2023. DOI.
15. **Han Zhou**, Wenjun Ying. A dimension splitting method for time dependent PDEs on irregular domains. *Journal of Scientific Computing*, 94(1):20, 2023. DOI.
16. Jiacheng Xu, Dan Hu, **Han Zhou**. A phase-field method for elastic mechanics with large deformation. *Journal of Computational Physics*, 471:111630, 2022. DOI.

TALKS

- 2026 SIAM Annual Meeting (AN26), Cleveland Jul. 6–10, 2026
- 2026 CAIMS/SCMAI Annual Meeting, Regina Jun. 15–19, 2026
- 2026 NCTS Workshop on Mathematics of Living Systems, Hsinchu May 20–22, 2026
- Kanazawa University - Penn Soft Matter / Applied Math Workshop, Philadelphia Mar. 23–25, 2026
- Mathematical Modeling, Computational Methods, and Biological Fluid Dynamics: Research and Training, Chicago Aug. 2025
- Numerical Analysis and PDE Seminar, University of Delaware, Newark, DE May 2025
- Fluid Mechanics and Waves Seminar, New Jersey Institute of Technology, Newark, NJ Nov. 2024
- NCTS Seminar on PDE and Machine Learning, Online Oct. 2024
- The 10th International Congress on Industrial and Applied Mathematics, Tokyo Aug. 2023
- SICIAM Workshop on Recent Advances in Fast Algorithms, Shenzhen Aug. 2023
- Chinese Mathematical Society in Computational Mathematics, Nanjing Jul. 2023
- CSIAM Student Forum, Online Nov. 2022
- Shanghai Symposium on Scientific and Engineering Computing Methods, Shanghai Nov. 2021

HONORS AND SCHOLARSHIPS

- Good Teaching Awards for Spring 2025 (UPenn math) 2025
- National Scholarship 2023
- Academic Scholarship for Graduate Students 2020-2022
- Excellent Bachelor Thesis of School of Mathematical Sciences 2020

TEACHING

University of Pennsylvania

- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, Spring 2026
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, Fall 2025
- **Instructor**, AMCS 6035, Numerical and Applied Analysis II, Spring 2025
- **Instructor**, AMCS 6025, Numerical and Applied Analysis I, Fall 2024

Shanghai Jiao Tong University

- **Teaching Assistant** Differential geometry (Fall 2023), Convex optimization (Spring 2022, Fall 2022), Scientific computing (Spring 2021, Fall 2021), Calculus (Fall 2020)

Graduate projects supervision(UPenn):

- Rui Xu (graduate student, AMCS) 2025
- JiYao Zhang (graduate student, AMCS) 2025

TECHNICAL SKILLS

Language: Mandarin, English

Computer programming C++, Python, MATLAB.

ACADEMIC SERVICE

Organizer, weekly UPenn AMCS Seminar

2025-2026

Organizer, weekly UPenn Mathematical Biology Seminar

2024-2025